

A Discretised Zeroing Neural Networks with Low Computational Complexity for Real Time Robotic Kinematic Control

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Abstract. Addressing the challenge of solving dynamic nonlinear equation systems (DNES) in robotic arm trajectory tracking, this paper extends the low computational complexity zeroing neural network (LCCZNN) through a discretized implementation for industrial control systems. Considering the point to point transmission characteristics of networked controllers, the continuous LCCZNN is discretized via the forward Euler method, yielding a discrete variant (DLCCZNN). Compared with higher order difference schemes, Euler discretization demonstrates superior numerical stability when tracking trajectories with dynamic singularities (e.g., heart shaped cusps), effectively avoiding computational overload. The influence of the activation function’s power parameter r on discrete time performance is also examined. Although LCCZNN theoretically guarantees finite time convergence, in practice, r critically affects stability and convergence quality. For $r < 1$ (e.g., 0.9), the system exhibits singular gain near the origin, enabling convergence to a bounded error in finite time. In the small error regime, where $r = 1$ behaves approximately linearly—the $r < 1$ case achieves significantly faster convergence. Finally, simulations and experiments on a UR3e manipulator in the planar plane validate the effectiveness of DLCCZNN. Results show that an appropriate balance between convergence gain μ and sampling interval τ ensures high accuracy tracking while preserving the low computational cost of LCCZNN.

Keywords: Zeroing neural network · LCCZNN · discretization · parameter analysis · finite time convergence · robotic arm control.

1 INTRODUCTION

The rapid advancement of industrial automation has established robotic manipulators as a cornerstone of modern manufacturing [1], expanding their role from basic material handling to complex, repetitive operations. In this context, robotic systems have evolved into key execution units for embodied intelligence [2]. Unlike traditional industrial settings, robotic arms within the embodied intelligence

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paradigm leverage multimodal perception to perform sophisticated tasks. A representative example is the integration of vision language models, which enables autonomous grasping and manipulation based on semantic instructions commonly referred to as open domain object interaction. Bridging the gap between theoretical models and real world deployment, however, requires achieving sub millimeter precision, which remains a fundamental challenge. Established control theory indicates that attaining such high precision performance necessitates the development of advanced algorithmic frameworks.

Recurrent Neural Dynamics (RND) has garnered significant attention as a robust framework for resolving intricate dynamic control challenges, favored for its superior nonlinear mapping and inherent parallel processing architecture. In tasks such as time varying parameter identification, RND frameworks facilitate continuous error rectification via state feedback loops. As a prominent derivative of RND, the Zeroing Neural Network (ZNN), pioneered by Zhang et al., exhibits exceptional efficacy in tracking time dependent linear and nonlinear systems. ZNN have gained significant prominence across a broad spectrum of conventional industrial applications, ranging from redundant manipulator control to real time trajectory tracking. Beyond conventional industrial applications, ZNN have demonstrated exceptional performance in specialized fields requiring stringent safety constraints, such as the kinematic control of surgical manipulators under Remote Center of Motion (RCM) requirements [3]. Unlike conventional Gradient Neural Networks (GNNs), the ZNN paradigm utilizes time derivative information to ensure exponential or finite time error decay [4, 5], effectively neutralizing latency induced tracking deviations. However, the physical implementation of classic ZNN models is often hindered by two primary bottlenecks: first, the heavy reliance on Jacobian pseudoinverse calculations leads to an algorithmic overhead of $O(mn^2)$, which restricts high frequency control. Second, most existing ZNN literature focuses on continuous time domains, whereas practical industrial controllers, such as the UR3e platform, operate on discrete time sampling.

To alleviate the first limitation, Zheng et al. [6] proposed the low computational complexity zeroing neural network (LCCZNN), which reformulates the control law through explicit dynamics and reduces the computational complexity to $O(mn)$. Recent advancements have extended the LCCZNN framework to address increasingly complex operational requirements, such as smooth time optimal motion control subject to hybrid joint constraints [7]. Beyond these functional expansions, various activation functions have been proposed to further improve the response speed of LCCZNN models. Notably, recent research has introduced enhanced LCCZNN formulations capable of achieving predefined time convergence for constrained time varying quadratic programming, providing a deterministic upper bound for the settling time regardless of initial error magnitudes [8]. However the remaining challenge is therefore to construct a discrete time counterpart that can be implemented on sampled data robotic systems without sacrificing the favorable convergence properties of LCCZNN. Although high order discretization schemes, such as Taylor expansion and Runge Kutta

methods, usually offer smaller truncation errors [9], their direct application to LCCZNN is not always desirable. Because LCCZNN exhibits fast nonlinear corrective dynamics, high order prediction can amplify numerical errors, especially near singular trajectory segments, and may therefore lead to instability in discrete implementation. For this reason, the forward Euler method is adopted in this study as a more robust compromise between numerical stability and computational simplicity [10, 11].

In addition to the discretization scheme, the activation function plays a central role in determining the behavior of the resulting discrete system. We therefore analyze the effect of the power parameter r in the nonlinear activation function introduced in [6]. Our results show that, although $r = 1$ behaves approximately linearly in the small error regime, choosing $r < 1$ yields a tighter bounded tracking error and faster convergence in the discrete implementation. Accordingly, this paper makes three main contributions: it derives a forward Euler based discrete LCCZNN model for robotic trajectory tracking, clarifies how the parameters r , μ , λ , and τ affect stability and accuracy, and validates the resulting controller through simulations and UR3e experiments on a heart shaped trajectory with a cusp.

2 Preliminaries and Problem Formulation

This section formulates the UR3e trajectory tracking task as a DNES, reviews the continuous time LCCZNN model used as the starting point of the study, and then introduces the forward Euler discretization adopted for implementation.

2.1 Problem Formulation

The manipulator considered in this study has six degrees of freedom. Let the Cartesian position of the end effector be denoted by $\mathbf{p}_a(t) \in \mathbb{R}^m$ ($m = 3$), and let the joint angle vector be denoted by $\boldsymbol{\theta}(t) \in \mathbb{R}^n$. The forward kinematic relationship can then be written as:

$$\mathbf{p}_a(t) = \psi(\boldsymbol{\theta}(t)) \quad (1)$$

where $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a nonlinear mapping function. Trajectory tracking requires the end effector position $\mathbf{p}_a(t)$ to follow a prescribed desired trajectory $\mathbf{p}_d(t)$. To quantify this objective, we define the Cartesian tracking error as:

$$\boldsymbol{\Psi}(t) = \mathbf{p}_a(t) - \mathbf{p}_d(t) = \psi(\boldsymbol{\theta}(t)) - \mathbf{p}_d(t) \quad (2)$$

The control objective is to determine a suitable joint trajectory $\boldsymbol{\theta}(t)$ such that $\boldsymbol{\Psi}(t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$. Since the desired joint vector is not given explicitly, the tracking problem can be recast as the solution of a dynamic nonlinear equation system (DNES). Differentiating the above equation with respect to time yields:

$$\dot{\boldsymbol{\Psi}}(t) = \frac{\partial \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \frac{d\boldsymbol{\theta}(t)}{dt} - \frac{d\mathbf{p}_d(t)}{dt} = \mathbf{J}_\psi(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}(t) - \dot{\mathbf{p}}_d(t) \quad (3)$$

where $J_\psi(\boldsymbol{\theta}) \in \mathbb{R}^{m \times n}$ is the manipulator Jacobian. Its six columns correspond to the six joints of the UR3e manipulator, whereas its three rows correspond to the Cartesian coordinates of the end effector. The vector $\dot{\boldsymbol{\theta}}(t)$ denotes the joint velocity.

2.2 Continuous Time LCCZNN Model

To solve the DNES defined above, we adopt the continuous time LCCZNN model proposed by Zheng et al. [6]. Within this framework, a scalar Lyapunov type energy function $v(t)$ is introduced to measure the norm of the tracking error $\boldsymbol{\Psi}(t)$, formulated as follows:

$$v(t) = \frac{1}{2} \|\boldsymbol{\Psi}(t)\|_2^2 = \frac{1}{2} \boldsymbol{\Psi}(t)^T \boldsymbol{\Psi}(t) \quad (4)$$

In classical ZNN design, the energy function is forced to decay according to a prescribed dynamics:

$$\dot{v}(t) = -\mu \phi(v(t)) \quad (5)$$

where $\mu > 0$ is the convergence gain and $\phi(\cdot)$ is the activation function. Furthermore, by applying the chain rule to the tracking error dynamics in (3), the time derivative $\dot{v}(t)$ can be expanded as:

$$\dot{v}(t) = \frac{\partial v}{\partial \boldsymbol{\Psi}} \frac{d\boldsymbol{\Psi}}{dt} = \boldsymbol{\Psi}(t)^T \dot{\boldsymbol{\Psi}}(t) = \boldsymbol{\Psi}(t)^T (\mathbf{J}_\psi(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}}(t) - \dot{\mathbf{p}}_d(t)) \quad (6)$$

Combining equations (5) and (6) gives:

$$\boldsymbol{\Psi}(t)^T \mathbf{J}_\psi(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}}(t) = \boldsymbol{\Psi}(t)^T \dot{\mathbf{p}}_d(t) - \mu \phi(v(t)) \quad (7)$$

Conventional ZNN formulations typically derive the control input through pseudoinverse based operations, which increases the computational cost. To avoid this burden, the continuous time LCCZNN rewrites the control law as:

$$\dot{\boldsymbol{\theta}}(t) = -\frac{\mathbf{J}_\psi^T \boldsymbol{\Psi}}{\|\mathbf{J}_\psi^T \boldsymbol{\Psi}\|_2^2 + \lambda} (-\boldsymbol{\Psi}^T \dot{\mathbf{p}}_d + \mu \phi(v)) \quad (8)$$

where $\lambda \geq 0$ is a regularization parameter introduced to avoid singularity in the denominator.

2.3 Forward Euler Discretization

Because industrial robot controllers operate in discrete time, the continuous time model cannot be implemented directly and must therefore be discretized. Although higher order difference schemes can reduce local truncation error, they are more sensitive to the fast nonlinear dynamics of LCCZNN and may amplify numerical errors near singular trajectory segments. Since the primary goal of this work is a robust sampled data implementation, the forward Euler method

is adopted as a numerically stable and computationally efficient discretization strategy [9–11].

Using a first order approximation, the derivative of $\theta(t)$ at the sampling instant t_k is written as:

$$\dot{\theta}(t_k) \approx \frac{\theta(t_{k+1}) - \theta(t_k)}{\tau} = \frac{\theta_{k+1} - \theta_k}{\tau} \quad (9)$$

where $\tau > 0$ denotes the constant sampling interval.

3 DLCCZNN Model Analysis

3.1 Derivation of the DLCCZNN Model

The objective of this section is to derive the discrete counterpart of the continuous time LCCZNN for UR3e trajectory tracking. Substituting the forward Euler approximation (9) into the continuous time control law (8) yields:

$$\frac{\theta_{k+1} - \theta_k}{\tau} = - \frac{\mathbf{J}_k^T \boldsymbol{\Psi}_k}{\|\mathbf{J}_k^T \boldsymbol{\Psi}_k\|_2^2 + \lambda} (-\boldsymbol{\Psi}_k^T \dot{\mathbf{p}}_{d,k} + \mu\phi(v_k)) \quad (10)$$

where all variables are sampled at the interval τ . Rearranging the above equation gives the DLCCZNN update law proposed in this paper:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \tau \left[\frac{\mathbf{J}_k^T \boldsymbol{\Psi}_k}{\|\mathbf{J}_k^T \boldsymbol{\Psi}_k\|_2^2 + \lambda} (-\boldsymbol{\Psi}_k^T \dot{\mathbf{p}}_{d,k} + \mu\phi(v_k)) \right] \quad (11)$$

The above equation constitutes the core discrete control law for the UR3e manipulator. With this update law established, the convergence and stability of the system are primarily governed by the activation function, specifically the power parameter r . Following [6], the continuous time LCCZNN employs the following nonlinear activation function:

$$\phi(z) = \text{sig}^r(z) \exp(|z|) \quad (12)$$

The convergence behavior of the system is highly sensitive to the value of r . According to [6], choosing $r > 1$ triggers initial divergence in the presence of large errors, whereas $r = 1$ yields only exponential convergence in the small error regime. By contrast, $r < 1$ preserves the finite time convergence property to a bounded error neighborhood, substantially accelerating convergence for small errors, while the exponential term concurrently provides strong corrective action for large errors. Because an excessively small r may cause numerical difficulties in the discrete model arising from near infinite gradients associated with negative power terms in its derivative [9], the subsequent experiments adopt $r = 0.9$ as the nominal setting.

4 Experimental Validation and Data Analysis

4.1 Experimental Setup

To evaluate the numerical robustness of the proposed DLCCZNN under discretization, a heart shaped trajectory with a cusp was selected as the tracking target [12]. Unlike smooth paths such as circles or ellipses, the cusp of the cardioid requires an abrupt change in the end effector velocity direction within a very short time interval. This feature makes the trajectory a demanding benchmark for discrete time control, because numerical errors are more likely to accumulate near the singular segment and induce jitter or divergence. Successful tracking of this path therefore provides a stringent test of both the discretization strategy and the activation function.

The desired trajectory is a standard symmetric cardioid constrained to the YZ plane to remain within the reachable workspace of the manipulator: $\mathbf{p}_d(t) = \mathbf{p}_{offset} + s \begin{bmatrix} 0 \\ 16 \sin^3(\omega t) \\ \mathcal{Z}(t) \end{bmatrix}$ m, where $\mathcal{Z}(t) = 13 \cos(\omega t) - 5 \cos(2\omega t) - 2 \cos(3\omega t) - \cos(4\omega t)$.

As illustrated in the simulation environment (Fig.1(a)), the experimental parameters are configured with a central $\mathbf{p}_{offset} = [0.3, 0, 0.4]^T$, a scaling coefficient $s=0.008$, and an angular velocity $\omega = \pi/5$ (corresponding to a period $T=10$ s). Prior to the execution, the joint space configuration was initialized to align precisely with the starting posture required for the designated trajectory as $\theta_0 = [2\pi/9, -389\pi/900, -2177\pi/3600, 53\pi/1440, 5\pi/18, 0]^T$.

4.2 Verification of Trajectory Tracking Performance

The first experiment was designed to verify the feasibility and tracking accuracy of the proposed DLCCZNN controller. The nominal parameters were selected as $\lambda = 0$, $\mu = 50$, $\tau = 0.001$, and $r = 0.9$, and the manipulator state was updated according to the discrete DLCCZNN law. Under these settings, the simulated end effector trajectory closely follows the desired cardioid, as shown in Fig.1(b), which confirms the high fidelity tracking capability of the discrete controller.

The same controller was then implemented on the physical UR3e platform. The experimental result in Fig.1(c) shows that the manipulator also reproduces the prescribed heart shaped trajectory in the real system. This agreement between simulation and hardware validates the practical implementability of the DLCCZNN controller and indicates that the discrete formulation remains effective under real world sampled data operation.

4.3 Effect of Parameters on Trajectory Tracking

During the experiments, three qualitatively distinct behaviors were observed: stable tracking, trajectory jitter, and divergence. To identify practical parameter

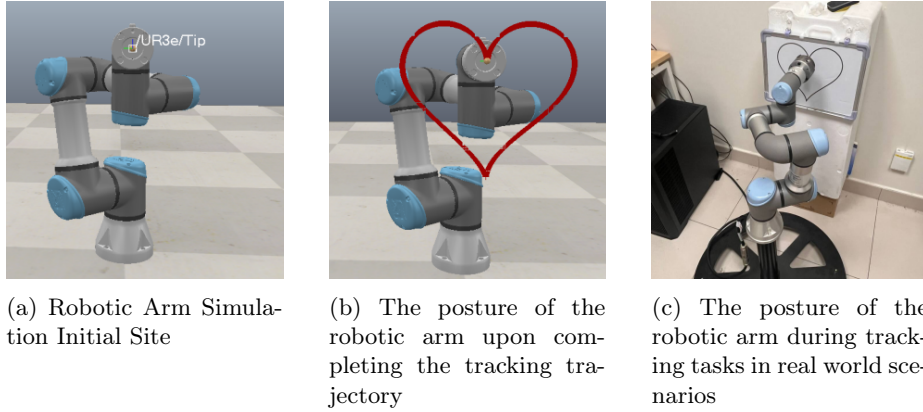


Fig. 1: Robotic Arm Simulation and Experiment Verification

ranges, a one factor at a time protocol was adopted, in which two parameters were fixed while the third parameter was varied.

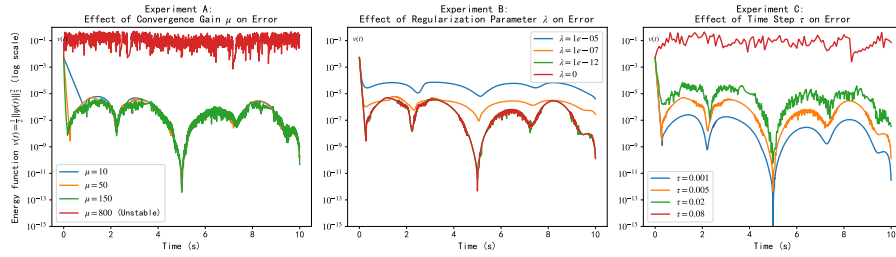
Fig.2(a) first illustrates the effect of the convergence gain μ . When μ increases from 10 to 50, the initial tracking error approaches the steady state error bound much faster, indicating that a larger convergence gain accelerates error decay [5]. However, once μ exceeds a critical threshold, the corrective action becomes excessively aggressive. For example, when $\mu = 800$, the controller overcompensates for the current error and drives the system to divergence, accompanied by severe oscillation of the manipulator. Within the stable range, however, varying μ has little influence on the final error bound, as reflected by the near overlap of the stable curves in Fig.2(a).

The regularization parameter λ is introduced to prevent singularity in the denominator of the control law [6]. As shown in Fig.2(a), increasing λ noticeably enlarges the steady state tracking error, which indicates that stronger regularization improves numerical safety at the cost of accuracy. From a practical standpoint, λ should therefore be kept as small as possible while still avoiding denominator singularity.

Fig.2(a) also shows that the sampling interval τ directly affects the achievable tracking accuracy. A smaller τ produces a tighter error bound because it reduces discretization error, but it also increases the number of iterations required over a fixed time horizon. Consequently, τ and λ must be coordinated carefully in implementation so that numerical robustness is maintained without unnecessarily sacrificing accuracy or efficiency [10].

4.4 Range Analysis of the Power Parameter r

Because the activation function contains a power term, the discrete time implementation imposes a practical lower bound on r for numerical stability. To



(a) The trajectory tracking error performances under different parameter settings.

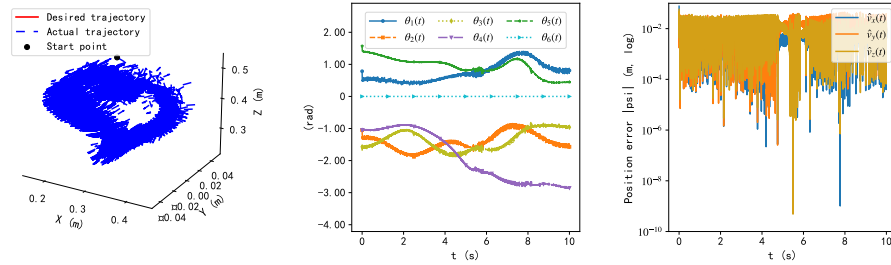
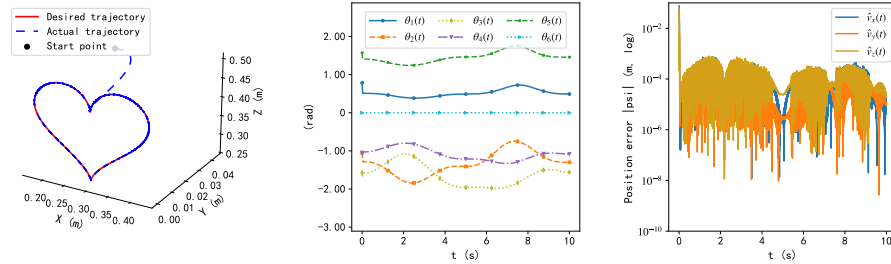
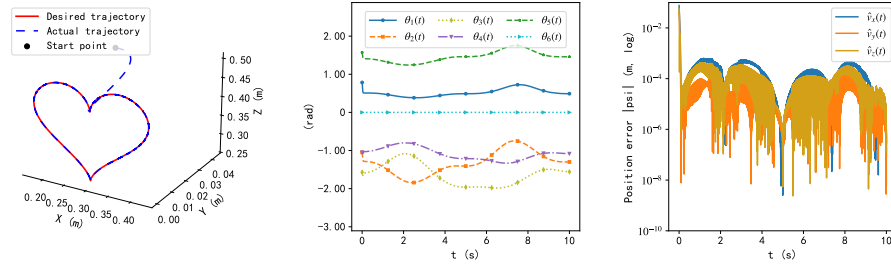
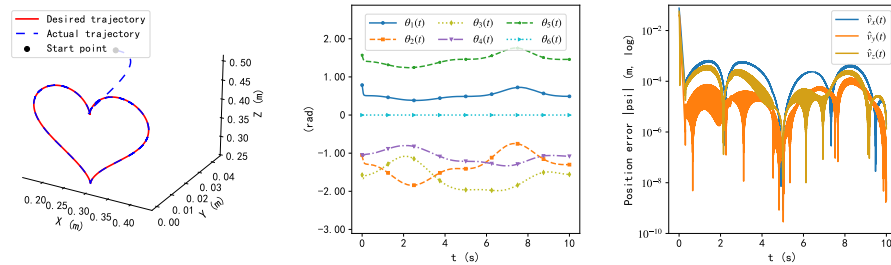
(b) The trajectory tracking, resolved joint angles, and positional error for $r = 0.5$.(c) The trajectory tracking, resolved joint angles, and positional error for $r = 0.8$.(d) The trajectory tracking, resolved joint angles, and positional error for $r = 0.9$.(e) The trajectory tracking, resolved joint angles, and positional error for $r = 1$.

Fig. 2: Influence of various parameters on computational convergence and error

determine an appropriate range, comparative simulations were carried out with $\lambda = 0$, $\mu = 50$, and $\tau = 0.001$, while r was varied.

When $r = 0.5$, the simulation exhibits severe numerical chattering, as shown in Fig.2(b), which indicates that excessively small values of r make the discrete controller overly sensitive and destabilize the tracking process. Increasing the parameter to $r = 0.8$ eliminates outright divergence, but noticeable high frequency oscillations remain, as shown in Fig.2(c), and the resulting tracking quality is still inadequate for precision control. By contrast, $r = 0.9$ produces a smooth trajectory fit without visible jitter, and the tracking error rapidly converges to a tight steady state bound, as shown in Fig.2(d). This result suggests that $r = 0.9$ provides a favorable balance between convergence speed and numerical stability in the discrete implementation.

For comparison, the near linear case $r = 1$ is shown in Fig.2(e). Although the system remains stable, the transient stage is substantially longer than that observed for $r < 1$, and the error fluctuation range is also broader. In other words, $r = 1$ loses the main practical advantage of the nonlinear activation function, namely rapid convergence to a tight bounded error neighborhood. For this reason, $r = 0.9$ is selected as the preferred value. Together with suitable choices of μ , λ , and τ , this setting confines the tracking error to a high precision range of approximately 0.01 mm to 1 mm.

5 Conclusion

This paper developed a discrete low computational complexity zeroing neural network (DLCCZNN) for real time robotic kinematic control and applied it to UR3e trajectory tracking. Starting from the continuous time LCCZNN formulation, a forward Euler based discrete update law was derived to enable sampled data implementation while preserving low computational complexity. Both simulation and hardware experiments demonstrate that, for the trajectory considered in this study, forward Euler discretization exhibits strong numerical robustness and effectively suppresses numerical instability near the singular segment of the heart shaped path.

The experiments further clarified the role of the activation function power parameter r in the discrete time model. In particular, choosing $r < 1$ (e.g., $r = 0.9$) preserves the singular gain effect near equilibrium and yields faster convergence to a bounded error neighborhood than the linear case ($r = 1$), whereas excessively small values of r induce numerical chattering. The parameter study also showed that high accuracy tracking depends on the coordinated tuning of the convergence gain μ , regularization parameter λ , and sampling interval τ . Under appropriate parameter settings, the proposed DLCCZNN achieves sub millimeter precision while retaining the computational efficiency of the original LCCZNN and avoiding pseudoinverse iterations.

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